

8. $0.1 = 10 \times 0.01$

9. Converting each of the given fractions into decimal form, we get:

$$\frac{3}{7} = 0.428, \frac{4}{11} = 0.363, \frac{5}{9} = 0.555, \frac{6}{13} = 0.461, \frac{7}{12} = 0.583$$

Clearly, $0.583 > 0.555 > 0.461 > 0.428 > 0.363$.

$$\text{So, } \frac{7}{12} > \frac{5}{9} > \frac{6}{13} > \frac{3}{7} > \frac{4}{11}.$$

10. Converting each of the given fractions into decimal form, we get:

$$\frac{2}{3} = 0.66, \frac{3}{5} = 0.6, \frac{7}{9} = 0.77, \frac{9}{11} = 0.81, \frac{8}{9} = 0.88.$$

Clearly, $0.6 < 0.66 < 0.77 < 0.81 < 0.88$.

$$\text{So, } \frac{3}{5} < \frac{2}{3} < \frac{7}{9} < \frac{9}{11} < \frac{8}{9}.$$

11. Converting each of the given fractions into decimal form, we get:

$$\frac{5}{9} = 0.55, \frac{7}{11} = 0.63, \frac{8}{15} = 0.533, \frac{11}{17} = 0.647.$$

Clearly, $0.647 > 0.63 > 0.55 > 0.533$.

$$\text{So, } \frac{11}{17} > \frac{7}{11} > \frac{5}{9} > \frac{8}{15}.$$

12. Converting each of the given fractions into decimal form, we get:

$$\frac{2}{3} = 0.66, \frac{3}{4} = 0.75, \frac{4}{5} = 0.8, \frac{5}{6} = 0.833.$$

Since $0.833 > 0.8 > 0.75 > 0.66$, so $\frac{5}{6} > \frac{4}{5} > \frac{3}{4} > \frac{2}{3}$.

$$\therefore \text{Required difference} = \left(\frac{5}{6} - \frac{2}{3} \right) = \frac{1}{6}.$$

13. Converting each of the given fractions into decimal form, we get:

$$\frac{15}{16} = 0.9375, \frac{19}{20} = 0.95, \frac{24}{25} = 0.96, \frac{34}{35} = 0.971$$

Clearly, $0.9375 < 0.95 < 0.96 < 0.971$. So, $\frac{15}{16} < \frac{19}{20} < \frac{24}{25} < \frac{34}{35}$.

14. Converting each of the given fractions into decimal form, we get:

$$\frac{9}{31} = 0.29, \frac{3}{17} = 0.176, \frac{6}{23} = 0.26, \frac{4}{11} = 0.363, \frac{7}{25} = 0.28.$$

Clearly, $0.363 > 0.29 > 0.28 > 0.26 > 0.176$.

$$\text{So, } \frac{4}{11} > \frac{9}{31} > \frac{7}{25} > \frac{6}{23} > \frac{3}{17}.$$

15. Converting each of the given fractions into decimal form, we get:

$$\frac{2}{5} = 0.4, \frac{3}{8} = 0.375, \frac{4}{9} = 0.444, \frac{5}{13} = 0.384, \frac{6}{11} = 0.545$$

Clearly, $0.375 < 0.384 < 0.4 < 0.444 < 0.545$.

$$\text{So, } \frac{3}{8} < \frac{5}{13} < \frac{2}{5} < \frac{4}{9} < \frac{6}{11}.$$

16. Clearly, $\frac{11}{14} = 0.785, \frac{16}{19} = 0.842, \frac{19}{21} = 0.904$.

Now, $0.785 < 0.842 < 0.904$. So, $\frac{11}{14} < \frac{16}{19} < \frac{19}{21}$.

17. We have:

$$\frac{13}{16} = 0.8125, \frac{15}{19} = 0.7894, \frac{17}{21} = 0.8095 \text{ and } \frac{7}{8} = 0.875.$$

Since 0.7894 is the smallest, so $\frac{15}{19}$ is the smallest.

18. $\frac{3}{4} = 0.75, \frac{5}{6} = 0.833, \frac{1}{2} = 0.5, \frac{2}{3} = 0.66, \frac{4}{5} = 0.8, \frac{9}{10} = 0.9$.

Clearly, 0.8 lies between 0.75 and 0.833.

$$\therefore \frac{4}{5} \text{ lies between } \frac{3}{4} \text{ and } \frac{5}{6}.$$

19. $\frac{7}{8} = 0.875, \frac{1}{3} = 0.333, \frac{1}{4} = 0.25, \frac{23}{24} = 0.958,$

$$\frac{11}{12} = 0.916, \frac{17}{24} = 0.708.$$

Clearly, 0.708 lies between 0.333 and 0.875.

$$\therefore \frac{17}{24} \text{ lies between } \frac{1}{3} \text{ and } \frac{7}{8}.$$

20. $\frac{4}{5} = 0.8, \frac{7}{13} = 0.53, \frac{1}{2} = 0.5, \frac{2}{3} = 0.66, \frac{3}{4} = 0.75, \frac{5}{7} = 0.714$.

Clearly, 0.5 does not lie between 0.53 and 0.8.

$$\therefore \frac{1}{2} \text{ does not lie between } \frac{4}{5} \text{ and } \frac{7}{13}.$$

21. $\frac{-7}{10} = -0.7, \frac{5}{-8} = -\frac{5}{8} = -0.625, \frac{2}{-3} = -\frac{2}{3} = -0.66$.

Since $-0.7 < -0.66 < -0.625$, so $\frac{-7}{10} < \frac{2}{-3} < \frac{5}{-8}$.

22. $\frac{2}{3} = 0.666, \frac{3}{5} = 0.6, \frac{2}{5} = 0.4, \frac{1}{3} = 0.333, \frac{1}{15} = 0.066, \frac{31}{50} = 0.62$.

Clearly, 0.62 lies between 0.6 and 0.666. So, $\frac{31}{50}$ lies

between $\frac{2}{3}$ and $\frac{3}{5}$.

23. $\frac{1}{5} = 0.2, \frac{8}{35} = 0.228, \frac{8}{37} = 0.216, \frac{8}{39} = 0.205, \frac{2}{11} = 0.181$.

Clearly, $0.181 < 0.2$. So, $\frac{2}{11} < \frac{1}{5}$.

24. $\frac{2}{5} = 0.4, \frac{4}{5} = 0.8, \frac{21}{50} = 0.42, \frac{20001}{50000} = 0.40002,$
 $\frac{200001}{500000} = 0.400002$.

Clearly, 0.400002 is nearest to 0.4.

$$\text{So, } \frac{200001}{500000} \text{ is nearest to } \frac{2}{5}.$$

25. $\frac{1}{3} = 0.333, \frac{3}{4} = 0.75, \frac{117}{300} = 0.39, \frac{287}{400} = 0.7175,$
 $\frac{95}{300} = 0.316, \frac{301}{400} = 0.7525, \frac{99}{300} = 0.33,$